

Light Pressure in PPC Law of Gravity

Light Pressure in Pawan Upadhyay's Pressure Curvature Law of Gravity (PPC Law of Gravity)

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Light Pressure in Pawan Upadhyay's Pressure–Curvature Law of Gravity

Light is often described as massless and intangible, yet it can push objects, move spacecraft, and influence matter across vast distances. This ability of light to exert force is known as **light pressure** or **radiation pressure**. While well established in physics, light pressure is often treated as a special case, separate from gravity.

In **Pawan Upadhyay's Pressure–Curvature Law of Gravity (PPC Law)**, light pressure fits naturally into a broader physical picture—one in which **pressure arises from energy density**, and pressure produces real physical effects.

What Is Light Pressure?

Light consists of photons, which have:

- zero rest mass,

- nonzero energy,

- nonzero momentum.

When photons strike a surface or are absorbed or reflected, they transfer momentum. This momentum transfer produces a force. When spread over an area, that force appears as **pressure**.

This phenomenon is experimentally verified and is the basis for:

- solar sails,
- radiation-driven dust motion,
- laser manipulation of small particles.

Pressure Without Mass

A key insight of the PPC framework is simple but powerful:

Pressure does not require mass; it requires energy density.

Photons have no rest mass, but they carry energy. That energy density naturally produces pressure. Light pressure therefore confirms a central idea of PPC gravity: **energy density alone is sufficient to generate pressure and force**.

Light Pressure as a Surface Force

In PPC terms, light pressure acts primarily as a **surface force**. When light strikes an object, the force depends on:

- the intensity of the light,
- the area of the surface,
- whether the light is absorbed or reflected.

This explains why:

- solar sails accelerate away from the Sun,
- comet tails point outward,
- thin reflective surfaces respond strongly to radiation.

Unlike gravity, light pressure does not usually act as a long-range field force; it acts locally where light interacts with matter.

Light and Spacetime Curvature

Light does not strongly curve spacetime in everyday situations, but it is not independent of curvature. In General Relativity, light follows **null geodesics**, meaning its path is determined by spacetime geometry.

In PPC gravity:

- gravitational pressure shapes spacetime curvature,
- curvature guides the motion of light,
- light therefore behaves as a **curvature-guided wave**.

This does not mean light is a gravitational wave. Light remains an electromagnetic wave, but its propagation reveals the structure of curved spacetime.

The Fastest Possible Speed

Another important point in PPC gravity is that light travels at the **fastest possible speed allowed by spacetime**. Because photons are massless, they move along the most direct causal paths. No signal, force, or pressure–curvature disturbance can propagate faster.

In PPC terms:

The speed of light is the maximum speed at which energy, information, and pressure-related effects can travel through spacetime.

Gravitational pressure–curvature waves also propagate at this same fundamental speed.

Light Pressure vs Gravitational Pressure

While both arise from energy density, light pressure and gravitational pressure are not the same:

- Light pressure acts mainly as a surface force through momentum transfer.
- Gravitational pressure acts as a field force that curves spacetime and governs motion.
- Light pressure produces mechanical effects; gravitational pressure produces geometry.

This distinction helps clarify why light can push objects without behaving like mass.

Why This Matters

Understanding light pressure through the PPC framework:

- unifies electromagnetic and gravitational phenomena conceptually,
- avoids outdated ideas like “relativistic mass,”
- highlights pressure as a universal physical agent,
- and deepens intuition about how energy interacts with spacetime.

Light pressure is not an exception to gravity—it is an example of how **energy density naturally produces pressure and force**, even without mass.

Final Thought

In Pawan Upadhyay's Pressure–Curvature Law of Gravity, light pressure demonstrates that pressure is a fundamental consequence of energy density, while gravity is the special case where pressure shapes spacetime itself.

— Pawan Upadhyay
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Pawan Upadhyay's Pressure–Curvature Law of Gravity